Predictable triangular mixed-Gram one-use theorem and source-graph Feshbach boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

R-139 changed the future problem from a sum of separated positive insertion cells to one complete signed terminal-minus-prefix trace-excess secant. This removes the R-138 reveal-to-insertion reanchoring factor, but it does not by itself estimate the resulting production response. The present checkpoint isolates the smallest analytic theorem that would estimate it without using the Cameron–Martin source more than once.

The first result is a predictable triangular coanalysis–Hardy theorem. It combines every future source block before taking the output square and then uses martingale orthogonality once, at the source endpoint. The production exponents already make the associated scalar triangular majorant Hilbert–Schmidt. Consequently exponent arithmetic is no longer the active obstruction. The missing production input is an owner-complete conditional mixed-Gram envelope for the actual R-102 whole-product response, including its product-covariance, trace, variance, forest, and mean/low coordinates inside one signed form.

The second result is an exact source-metric graph–Feshbach identity. It keeps the R-131 canonical source coordinate outside the physical pullback, retains the low-block kernel defect when the low diagonal is semidefinite, and gives a sharp scalar norm-data source-gap criterion. It is deliberately weaker than R-139 ambient positivity: the desired gap is measured only in the source norm on the actual production graph.

Everything here is finite-cutoff, fixed-floor, fixed-chart, and conditional on the displayed operator hypotheses. No cutoff-, chart-, cluster-, visit-, control-, or refinement-uniform production mixed-Gram estimate is proved. The old number K_0 is used only as a conditional diagnostic and is not a proved bound for the production cross operator. No complete low theorem, matching theorem, strict production gap, absolute anchor, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson bound, removal, interacting measure, Sector-A closure, or tier promotion follows.

2. Predictable future-source coordinate

Fix a finite filtration $(\mathcal{F}_r)_{r=\ell}^{N}$ and finite-dimensional Hilbert spaces H_k and Y_m. Let

$$h_k \in L^2(\mathcal{F}_{k-1}; H_k), \qquad k > \ell, \quad d_r = P_r - P_{r-1}. \tag{2.1}$$

For every reveal r, collect all still-future source innovations in

$$x_r = (d_r h_k)_{k>r} \in \bigoplus_{k>r} H_k. \tag{2.2}$$

Let $T_{r,m,k} : H_k \to Y_m$ be $\mathcal{F}_{r-1}$-predictable. The load-bearing production factorisation sought after R-139 is

$$d_r\big(J^N - J^{n(r)}\big)_m = \sum_{k>r} T_{r,m,k} d_r h_k. \tag{2.3}$$

The equality is an owner-complete statement: the response on its left must already contain the same-root recombination and the whole R-102 product residual. Factorwise paraproduct identities are not substitutes for (2.3).

For collar $C \ge 5$ and weight exponent $\gamma > 0$, define

$$(A_r^C x)_m = \mathbf{1}_{\{m \ge r+C\}} 2^{\gamma(m-r-C)} \sum_{k>r} T_{r,m,k} x_k. \tag{2.4}$$

The norm is from the future-source direct sum to the output-shell direct sum.

3. The predictable triangular one-use theorem

Theorem 3.1. Assume (2.3) and

$$\operatorname*{ess\,sup}_{r,\omega} \|A_r^C(\omega)\| \le \Lambda_C. \tag{3.1}$$

Then

$$\begin{aligned} \sum_r \sum_{m \ge r+C} & 2^{2\gamma(m-r-C)} \mathbb{E} \left\| \sum_{k>r} T_{r,m,k} d_r h_k \right\|^2 \\ & \le \Lambda_C^2 \sum_r \sum_{k>r} \mathbb{E}\|d_r h_k\|^2 \\ & = \Lambda_C^2 \sum_k \mathbb{E}\|h_k - P_\ell h_k\|^2 \le \Lambda_C^2 X_{\mathrm{src}}. \end{aligned} \tag{3.2}$$

Proof. Apply (3.1) pointwise to x_r, integrate, and sum over r. Exchange the two finite source sums. Since h_k is $\mathcal{F}_{k-1}$-measurable, the increments $d_r h_k$, $\ell < r < k$, are orthogonal; therefore

$$\sum_{\ell<r<k} \mathbb{E}\|d_r h_k\|^2 = \mathbb{E}\|h_k - P_\ell h_k\|^2. \tag{3.3}$$

This is the only Doob-square use. No insertionwise positive square and no factor $2^{2\gamma(k-r)}$ is introduced. □

The same proof applies directly to the full signed trace-excess when it has an owner-complete quadratic representation

$$S_C = \sum_r \mathbb{E}\langle x_r, Q_r^C x_r\rangle, \qquad Q_r^C \preceq \Lambda_C^2 \mathrm{I}. \tag{3.4}$$

Then $S_C \le \Lambda_C^2 X_{\mathrm{src}}$ and R-139 gives

$$\mathfrak{I}_{\mathrm{fut}}^C \ge -\frac{\Lambda_C^2}{2} X_{\mathrm{src}}. \tag{3.5}$$

Thus Λ_C^2 is an adverse source-Hessian debt, not a new positive reserve. The radial Taylor action receives its one-half, not a second copy.

4. Exact scalar triangular majorant

Put $a = k - r \ge 1$, $d = m - r - C \ge 0$, $\delta = C - 5$, and suppose the block envelope has the production exponents

$$\|T_{r,m,k}\| \le C_* 2^{-\beta a/2} \begin{cases} 2^{-s(d-a+\delta)}, & d \ge a - \delta, \\ 1, & d < a - \delta. \end{cases} \tag{4.1}$$

The harmless translation from $2^{-\beta k/2}$ to $2^{-\beta(k-r)/2}$ is part of the normalized source-block hypothesis; production must prove it, not infer it from this scalar calculation. Set

$$q = 2^{2\gamma}, \quad u = 2^{-(\beta-2\gamma)}, \quad v = 2^{-\beta}, \quad \rho = 2^{-2(s-\gamma)}, \quad z = 2^{-(\beta-2s)}. \tag{4.2}$$

When $\beta/2 > \gamma$ and $s > \gamma$, the squared Hilbert–Schmidt majorant $H_C = \|A_r^C\|_{\mathrm{HS}}^2/C_*^2$ is finite. At $C = 5$ it is exactly

$$\boxed{H_5 = \frac{u/(1-u) - v/(1-v)}{q-1} + \frac{u}{(1-u)(1-\rho)}.} \tag{4.3}$$

For $\delta \geq 0$ the exact decomposition into near, newly-far low columns, and far high columns is

$$\begin{aligned} H_C = {} & \frac{q^{-\delta}u^{\delta+1}/(1-u) - v^{\delta+1}/(1-v)}{q-1} \\ & + \frac{2^{-2s\delta}}{1-\rho}\sum_{a=1}^{\delta} z^a + \frac{2^{-2\gamma\delta}u^{\delta+1}}{(1-\rho)(1-u)}. \end{aligned} \tag{4.4}$$

The middle sum is zero for $\delta = 0$. Formula (4.4) follows by summing $d < a - \delta$, then $a \leq \delta$, then $a > \delta$; it is not a fitted constant.

For the registered values

$$\beta = \frac{7}{5}, \qquad s = \frac{2}{3}, \qquad \gamma = \frac{7}{12}, \tag{4.5}$$

the two strict margins are

$$\frac{\beta}{2} - \gamma = \frac{7}{60}, \qquad s - \gamma = \frac{1}{12}. \tag{4.6}$$

The executable evaluates $H_5 = 56.2981540\ldots$ and $H_6 = 24.806198\ldots$ independently from finite double sums. If, only as a diagnostic, one sets $C_* = 1$ and multiplies by the R-084/R-138 normalized six-row coefficient $3/(40P)$, $P = 4 + 10^{-12}$, the action half-debts are approximately 0.5278 at $C = 5$ and 0.2326 at $C = 6$. These numbers do not certify the production C_*, the whole mixed Gram, or either A13 gate. In fact the corresponding source-Hessian debts are $1.05559\ldots$ and $0.465116\ldots$, both larger than the zero-low/tail conditional headroom $0.100434\ldots$ in Section 9. Thus the naive positive current-only scalar majorant is summable but quantitatively too coarse even at $C = 6$. A smaller production mixed-Gram constant or signed trace/variance/forest cancellation is indispensable.

5. The R-102 whole-product firewall

For bilinear multiplication $m(H, c) = Hc$, R-102 proves

$$\begin{aligned} d_r m(H,c) = {} & m(d_r H, P_{r-1}c) + m(P_{r-1}H, d_r c) + m(d_r H, d_r c) \\ & + \mathrm{Cov}_r^m(H,c) - \mathrm{Cov}_{r-1}^m(H,c). \end{aligned} \tag{5.1}$$

The first two terms are predictable paraproducts and are compatible with Theorem 3.1 after their production block bounds are proved. The last line together with $m(d_r H, d_r c)$ is a signed innovation. It must remain with the primitive trace, R-125 future variance, R-063 forest coordinate, and predictable mean/low endpoint inside the single form Q_r^C of (3.4). Paying it by a second independent Λ^2 estimate would duplicate an owner. R-087 expectation-wise atom bounds do not yet imply this conditional owner-complete mixed-Gram statement.

This identifies the exact production gap:

$$\boxed{\begin{array}{l}\text{prove (2.3) or (3.4) for the complete R-102 product residual,}\\ \text{uniformly in the declared production parameters.}\end{array}} \tag{5.2}$$

The strict exponent margins (4.6) show that the scalar summation of such an envelope is available; they do not construct the envelope.

6. Predictable stopping preserves the endpoint telescope

Let $\widehat{K}^q_{r,m}$ be the R-139 conditional endpoint energy and let $\widehat{\omega}_{e,r,m} = \widehat{K}^e_{r,m} - \widehat{K}^{e-1}_{r,m}$. If $\sigma_{r,m}$ is bounded and $\mathcal{F}_{r-1}$-predictable, and the same multiplier is used for every future insertion $e > n(r)$, then

$$\sum_{e>n(r)} \mathbb{E}[\sigma_{r,m}\widehat{\omega}_{e,r,m}] = \mathbb{E}[\sigma_{r,m}(\widehat{K}^N_{r,m} - \widehat{K}^{n(r)}_{r,m})]. \tag{6.1}$$

In particular, a predictable stopped multiplier such as $\sigma_{r,m} = \mathbf{1}_{\{\tau_m \geq r\}}$ is legal. This does not contradict the R-139 wedge no-go, which concerns an insertion-dependent mask χ_e. Stopping alone creates no good-cell payment; a source/sextic Carleson or BMO packing theorem for the stopped bad set is still absent.

7. Exact source-metric graph–Feshbach theorem

Let $\mathcal{S}, \mathcal{Y}, \mathcal{L}$ be finite-dimensional Hilbert spaces. Let E, F be self-adjoint, $D \succeq 0$, and let $A_0 : \mathcal{Y} \to \mathcal{S}$, $B : \mathcal{L} \to \mathcal{S}$, $C_0 : \mathcal{L} \to \mathcal{Y}$. Write

$$M_2 = \begin{pmatrix} E & -A_0/2 \\ -A_0^*/2 & F \end{pmatrix}, \qquad K = \begin{pmatrix} B \\ C_0 \end{pmatrix}. \tag{7.1}$$

For production tangents $V : \mathcal{S} \to \mathcal{Y}$ and $\Lambda : \mathcal{S} \to \mathcal{L}$, put

$$Zu = (u, Vu), \qquad \widehat{G}u = (u, Vu, \Lambda u). \tag{7.2}$$

Let P_0 project onto $\operatorname{Ker} D$, $P_+ = \mathrm{I} - P_0$, and let W be the reduced solution

$$D^{1/2}W = P_+K^*Z. \tag{7.3}$$

Define

$$N = P_0K^*Z, \qquad \Delta = D^{1/2}\Lambda - W, \qquad \Lambda_0 = P_0\Lambda. \tag{7.4}$$

Theorem 7.1. For

$$\mathcal{M}(T) = \begin{pmatrix} E & -(A_0+T)/2 & -B \\ -(A_0+T)^*/2 & F & -C_0 \\ -B^* & -C_0^* & D \end{pmatrix}, \tag{7.5}$$

one has the exact zero-tail identity

$$\begin{aligned} \langle \widehat{G}u, \mathcal{M}(0)\widehat{G}u \rangle = \langle Zu, M_2Zu \rangle - \|Wu\|^2 + \|\Delta u\|^2 \\ - 2\operatorname{Re}\langle Nu, \Lambda_0 u \rangle. \end{aligned} \tag{7.6}$$

Consequently, for every $T : \mathcal{Y} \to \mathcal{S}$ with $\|T\| \leq \tau$,

$$\langle \widehat{G}u, \mathcal{M}(T)\widehat{G}u \rangle \geq \mu\|u\|^2 \tag{7.7}$$

for every u if and only if the right side of (7.6) satisfies

$$\langle Zu, M_2Zu \rangle - \|Wu\|^2 + \|\Delta u\|^2 - 2\operatorname{Re}\langle Nu, \Lambda_0 u \rangle \geq \tau\|u\|\|Vu\| + \mu\|u\|^2. \tag{7.8}$$

The product term is sharp, attained pointwise by an aligned rank-one tail.

Proof. Expand the low cross as $-2\operatorname{Re}\langle K^*Zu, \Lambda u \rangle + \|D^{1/2}\Lambda u\|^2$. Complete the square on $P_+\mathcal{L}$ and retain the $P_0\mathcal{L}$ cross. This gives (7.6). The tail contribution is $-\operatorname{Re}\langle u, TVu \rangle$; optimizing it over the operator-norm ball gives exactly $-\tau\|u\|\|Vu\|$. ☐

If the graph-Douglas compatibility

$$K^*Z(\mathcal{S}) \subseteq \operatorname{Ran} D^{1/2} \tag{7.9}$$

holds, then $N = 0$. Without (7.9), deleting the last term of (7.6) is false. The executable includes a semidefinite fixture in which the true value is -15 while the formula with the kernel cross omitted gives 15.

8. Sharp scalar source-gap corollary

Assume now

$$D \succeq d\mathrm{I}, \quad E \succeq e\mathrm{I}, \quad F \succeq f\mathrm{I}, \quad \|K\| \le k, \quad \|A_0\| \le a_0, \quad \sigma = \frac{k^2}{d}. \tag{8.1}$$

If $f > \sigma$, then (7.8) is implied by

$$\boxed{\mu_{\mathrm{src}} = e - \sigma - \frac{(a_0+\tau)^2}{4(f-\sigma)} > 0.} \tag{8.2}$$

Indeed, discard $\|\Delta u\|^2$ and use

$$(e-\sigma)\|u\|^2 + (f-\sigma)\|Vu\|^2 - (a_0+\tau)\|u\|\|Vu\|. \tag{8.3}$$

Minimization in $\|Vu\|/\|u\|$ gives (8.2). Equivalently,

$$a_0 + \tau < 2\sqrt{(e-\sigma)(f-\sigma)}. \tag{8.4}$$

For a prescribed source gap μ, only the source diagonal is shifted:

$$a_0 + \tau \le 2\sqrt{(e-\sigma-\mu)(f-\sigma)}. \tag{8.5}$$

Shifting F or D by μ would demand the stronger ambient gap and is not equivalent to (7.7). The elementary identity

$$4x^2 + 9y^2 - 6xy = 3x^2 + (3y-x)^2 \tag{8.6}$$

has source gap 3 but fails after subtracting 3 from both diagonals.

Within the norm-data class, (8.2) is sharp. Take scalar aligned A_0, T, choose K and D so the Schur loss is σ, put

$$Vu = \frac{a_0+\tau}{2(f-\sigma)} u, \tag{8.7}$$

and choose Λu as the low Schur minimizer. Equality is attained; at $\mu_{\mathrm{src}} = 0$ there is a kernel. This is a sharpness fixture, not a counterexample to a production estimate with stronger structural data.

9. Once-owned debt composition and diagnostics

R-131 pins the source Hessian as

$$D^2\mathcal{A}_J(h) = \frac{9}{10}\mathrm{I} + L_{\mathrm{pred}}^*(Q_{\mathrm{comp}} + Q_6)L_{\mathrm{pred}}. \tag{9.1}$$

Thus a source-metric estimate is the correct weaker target. Reusing the R-128 half-debt diagnostic, let

$$P = 4 + 10^{-12}, \quad \eta_h = \frac{9}{20} - \frac{3}{125P} - \frac{1}{880}, \quad \zeta_h = \frac{27}{200}, \tag{9.2}$$

$$e = 2\eta_h, \qquad f = 2\zeta_h, \qquad K_0 = 4\sqrt{\left(\frac{197}{440} - \frac{3}{125P}\right)\frac{3}{25}}. \tag{9.3}$$

Here e, f are valid only after the declared diagonal debts have already been absorbed. Also $a_0 = K_0$ below is a conditional test input, not a proved production inequality.

At $\sigma = \tau = 0$, (8.2) gives

$$\mu_{\mathrm{src}} = 0.1004343434\ldots. \tag{9.4}$$

The zero-tail source-gap equation has its first low-loss root at $\sigma = 0.0239601163\ldots$. Taking half that diagnostic loss and the test tail $\tau = 0.02$ leaves

$$\mu_{\mathrm{src}} = 0.0159126895\ldots. \tag{9.5}$$

These are headroom diagnostics only. The production values of a_0, d, k, τ and all residual diagonal forms remain unproved.

Combining Theorem 3.1 with (8.2) uses the future debt exactly once:

$$\boxed{\mu_{\mathrm{src}}(C) = e - \Lambda_C^2 - \sigma - \frac{(a_0 + \tau_C)^2}{4(f - \sigma)}.} \tag{9.6}$$

Thus the zero-low/tail diagnostic would require $\Lambda_C^2 < 0.1004343434\ldots$, whereas the illustrative loss/tail point would require $\Lambda_C^2 < 0.0159126895\ldots$. Neither inequality is currently a production theorem. Moreover, μ_{src} is Hessian slack; radial Taylor contributes only $\mu_{\mathrm{src}}\|h\|^2/2$ to the action. The additive R-123 constant is separate anchor bookkeeping and never enters (7.6)–(9.6).

10. Ownership, hostile fixtures, and rejected shortcuts

The verification package checks all of the following independently.

1. $R^*F = F^*R = H$ for a self-adjoint endpoint owner, while adding both directions gives $2H$. The Hessian convention takes the half-symmetrization and owns H once.
2. A positive-D rational fixture gives the same direct and completed- square value, 62.
3. A semidefinite-D fixture gives -15 only when the kernel cross in (7.6) is retained.
4. The source-only shift fixture (8.6) rejects an accidental ambient shift.
5. An aligned rank-one tail attains the product in (7.8), and the scalar Schur minimizer attains (8.2).
6. Finite triangular sums converge to (4.3)–(4.4), so the strict margins in (4.6), rather than rounded decimal output, drive convergence.

Several routes are therefore parked rather than silently forgotten. Insertion-dependent wedge stopping still leaves the R-139 internal variation sum. A predictable stopping time preserves the telescope but supplies no bad-set packing. Estimating only the two predictable terms in (5.1) deletes the product covariance. Estimating that residual separately pays the same owner twice. Shifting all graph channels by a source gap proves a different, stronger statement. Finally, an absolute diagonal spatial envelope may discard the stationary current–trace cancellation; it cannot replace the signed complete mixed Gram unless its ultraviolet source sum is proved.

10.1 Absolute-envelope hostile audit and signed repair

The last warning can be made exact. Exhaustive R-125 output coanalysis and the R-067 injection convention give, at zero control,

$$\sum_m \mathbb{E}\Theta^0_{r,m} = 2\mathbb{E}\mathcal{I}^0_r. \tag{10.1}$$

R-067 proves a scale-zero shell injection of order one whose terminal raw energy is cancelled only in the signed stationary energy–injection packet. Therefore an absolute terminal envelope satisfies

$$Q^{\Theta,0}_{\sigma,J} = \sum_{r,m} 2^{2\sigma m}\mathbb{E}\Theta^0_{r,m} \geq 2\sum_{r=r_0}^{J} \mathbb{E}\mathcal{I}^0_r \gtrsim J - r_0 + 1. \tag{10.2}$$

In the scale-local diagonal fixture $m = r + C + 3$ with $\Theta_{r,m} = \theta_0 > 0$, it grows exactly as

$$Q^{\Theta,0}_{\sigma,J} = \theta_0\, 2^{2\sigma(C+3)} \frac{2^{2\sigma(J+1)} - 1}{2^{2\sigma} - 1}. \tag{10.3}$$

At $\sigma = 2/3$ this is exponential in J. Thus the previously considered absolute terminal H^σ envelope is production-infeasible unless a new subtraction is retained. This is covered by existing stationary/absolute-value boundaries and does not warrant a duplicate negative-result ID.

There is a valid signed repair. On a common heat/fresh-trace lift write $Z^q = (U^q, \Phi^q)$ with $\|U^q\|^2 = \Theta^q$ and $S(U,V) = (U, -V)$. For the common zero-control endpoint $Z^{\rm st}$ put $\widetilde{Z}^q = Z^q - Z^{\rm st}$. The secant polarizes exactly as

$$\begin{aligned} \mathcal{D}_C &= 2\,\mathrm{Re}\langle Z^{\rm st}, S(\widetilde{Z}^* - \widetilde{Z}^p)\rangle_w \\ &\quad + \mathrm{Re}\langle \widetilde{Z}^* + \widetilde{Z}^p, S(\widetilde{Z}^* - \widetilde{Z}^p)\rangle_w. \end{aligned} \tag{10.4}$$

For arbitrary $\rho_{r,m} > 0$, define

$$\begin{gathered} A_{\rm st} = \sum w\rho^{-1}\mathbb{E}\|Z^{\rm st}\|^2, \quad A_{\rm sec} = \sum w\rho\mathbb{E}\|\widetilde{Z}^* - \widetilde{Z}^p\|^2, \\ A_{\rm ctl} = \sum w\mathbb{E}(\|\widetilde{Z}^*\|^2 + \|\widetilde{Z}^p\|^2). \end{gathered} \tag{10.5}$$

Cauchy–Schwarz and the parallelogram identity give

$$\boxed{\mathcal{D}_C \le 2\sqrt{A_{\rm st}A_{\rm sec}} + A_{\rm ctl}} \le \varepsilon A_{\rm sec} + \varepsilon^{-1}A_{\rm st} + A_{\rm ctl}. \tag{10.6}$$

Choosing $\rho_{r,m} = 2^{2\sigma r}$ makes a scale-zero stationary shell summable, while the positive regularity weight falls on a secant that vanishes at zero control. However, production bounds for the weighted stationary FAR norm and the control secant remain open, so (10.6) is a route repair rather than a closure.

The hostile audit also corrects an earlier abstract fixture: taking $P_r = P_{r-1} = I$ forces the centered reveal innovation to vanish and is not a nontrivial R-139 owner. A filtration-consistent algebra fixture uses independent Rademacher variables ξ_r, $m_r = r{+}C{+}3$, $J^{p(r)} = \xi_r e_{m_r}$, $J^* = 0$, and $\Theta = 0$. Then $b = 0$, $Y = \xi_r e_{m_r}$, $D_0^{p(r)} = -1$, $D_0^* = 0$, and the weighted sum is $M2^{7/2}$. It deliberately omits common-terminal contraction and therefore tests the algebra only; it is not a production counterexample.

11. What has advanced and what remains open

The durable advance is a non-circular division of labour.

1. Production analysis must prove one predictable owner-complete mixed-Gram operator bound $Q_r^C \preceq \Lambda_C^2 \mathrm{I}$, including the R-102 whole-product residual and all trace/variance/forest/mean-low owners once.
2. The scalar shell sum is then automatic from (4.4); its registered exponents have strict margins.
3. The resulting Λ_C^2 is subtracted once from the source diagonal in (9.6).
4. Low and tail data enter the exact graph–Feshbach identity (7.6) and the source-gap test (7.8), not a duplicated ambient budget.
5. Only after a positive production value of (9.6) is certified may the R-136 radial Taylor route address matching and the absolute anchor.

The smallest successor remains inside A13 and T-050: prove the production conditional mixed-Gram envelope (5.2), with explicit uniform C_* and low-kernel compatibility, then evaluate (9.6). If it fails, the exact failure must identify which signed residual or graph parameter saturates the budget. No new claim card is warranted.

12. Devil's-advocate audit

1. **Objection:** Hilbert–Schmidt summability proves the production operator bound. **UPHELD as an overclaim.** It sums an assumed block envelope and does not construct that envelope.

2. **Objection:** R-087 already supplies the required conditional mixed Gram. **UPHELD as an overclaim.** Its expectation-wise atoms do not retain the whole R-102 product covariance inside one predictable form.

3. **Objection:** predictability is cosmetic. **DISMISSED.** It is the firewall preventing an anticipative response from adapting to $d_r h_k$ before the one-use source orthogonality is applied.

4. **Objection:** predictable stopping closes bad cells. **UPHELD as an overclaim.** Equation (6.1) preserves cancellation but proves no Carleson/BMO packing.

5. **Objection:** the low kernel term can be omitted by continuity. **DISMISSED.** The exact semidefinite fixture changes the value from -15 to 15 if it is deleted.

6. **Objection:** a source gap shifts every diagonal. **DISMISSED.** Equation (8.6) is a counterfixture; only E is shifted for the source-norm statement.

7. **Objection:** K_0 is the production a_0. **UPHELD as an overclaim.** It is an old acceptance ceiling used only to measure conditional headroom.

8. **Objection:** positive μ_{src} proves a Nelson/action margin. **UPHELD as an overclaim.** Radial integration halves the Hessian slack, while matching and the absolute anchor remain open.

9. **Objection:** the R-123 additive constant can enlarge Feshbach headroom. **DISMISSED.** An additive anchor constant does not alter a homogeneous quadratic operator inequality.

10. **Objection:** the two strict exponent margins close T-050. **UPHELD as an overclaim.** The production mixed-Gram constant and both A13 gates are still missing.

13. Result footer and reproduction

Primary certificate:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  (Get-Item codes/foundations/*mixed_gram_source_graph_feshbach_boundary.py)
```

Independent certificate:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  (Get-Item codes/foundations/*mixed_gram*boundary_independent.py)
```

Integrated verification:

```
E:\Dev\TECT.venv\Scripts\python.exe `
  (Get-Item codes/foundations/*mixed_gram*boundary_verify.py)
```

Result ID: `A13-CLASSII-PREDICTABLE-TRIANGULAR-MIXED-GRAM-SOURCE-GRAPH-FESHBACH-BOUNDARY`. Ledger ID: R-140. Tier remains T4.

No-overclaim statement. R-140 proves a conditional one-use triangular theorem, exact scalar summability, a predictable stopping identity, and exact source-graph Feshbach identities and sharp norm-data criteria. It does not prove the production mixed-Gram hypothesis, any uniform production constant, a positive production graph gap, matching, anchor, either A13 gate, Nelson, removal, interacting measure, or Sector-A closure.

Result ID: A13-CLASSII-PREDICTABLE-TRIANGULAR-MIXED-GRAM-SOURCE-GRAPH-FESHBACH-BOUNDARY.
Ledger ID: R-140.
Precise statement: Theorems 3.1 and 7.1; exact scalar sums (4.3)-(4.6); predictable stopping identity (6.1); source-gap criterion (8.2); once-debited formula (9.6); stationary subtraction (10.4)-(10.6).
Scope: fixed finite cutoff and positive floor; fixed temporally faithful chart; finite-dimensional Hilbert spaces; displayed predictability, factorization, operator-envelope, graph, and low-range hypotheses only.
Dependencies: A1; R-063, R-067, R-079, R-087, R-099, R-102, R-103, R-104, R-123, R-125, R-128, R-129, R-131, R-133, R-136, R-139.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_predictable_triangular_mixed_gram_source_graph_feshbach_boundary_verify.py
Expected output: R-140 integrated PASS; primary 65/65, independent 69/69, note/PDF, manifest, exploration, authority, and public gates pass.
Falsification gate: any failed one-use identity, scalar geometric sum, normalization or headroom comparison, stopping telescope, positive or semidefinite Feshbach fixture, source-only shift, sharpness fixture, stationary subtraction, authority/hash, PDF, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no production owner-complete mixed-Gram bound, uniform Lambda_C, low compatibility, positive source-graph margin, matching, absolute anchor, A13 gate, OVERLAP_src, Nelson estimate, removal, interacting measure, Sector-A closure, or T5-T7 claim follows.
Next required action: prove or falsify the complete R-102 predictable mixed-Gram envelope and production low compatibility, then insert only proved constants in (9.6) before matching, radial Taylor, and the anchor.